

1

a

Entity Sets

- Account
- Branch
- Customer
- Loan

Relations

- Account branch

Many Accounts to One Branch with all Accounts needing to have a branch

- Loan branch

Many Loans to One Branch with all Loans needing to have a branch

- Depositor

Accounts to Customer

- Borrower

Customer to Loan

b

Entities

Account(Account_No, Balance, B_name)

Branch(B_name, B_city, Assets)

Customer(C_name, C_city, C_street)

Loan(Loan_No, Amount, B_name)

Relations

Depositor(Account_No, C_name)

Borrower(C_name, Loan_No)

c

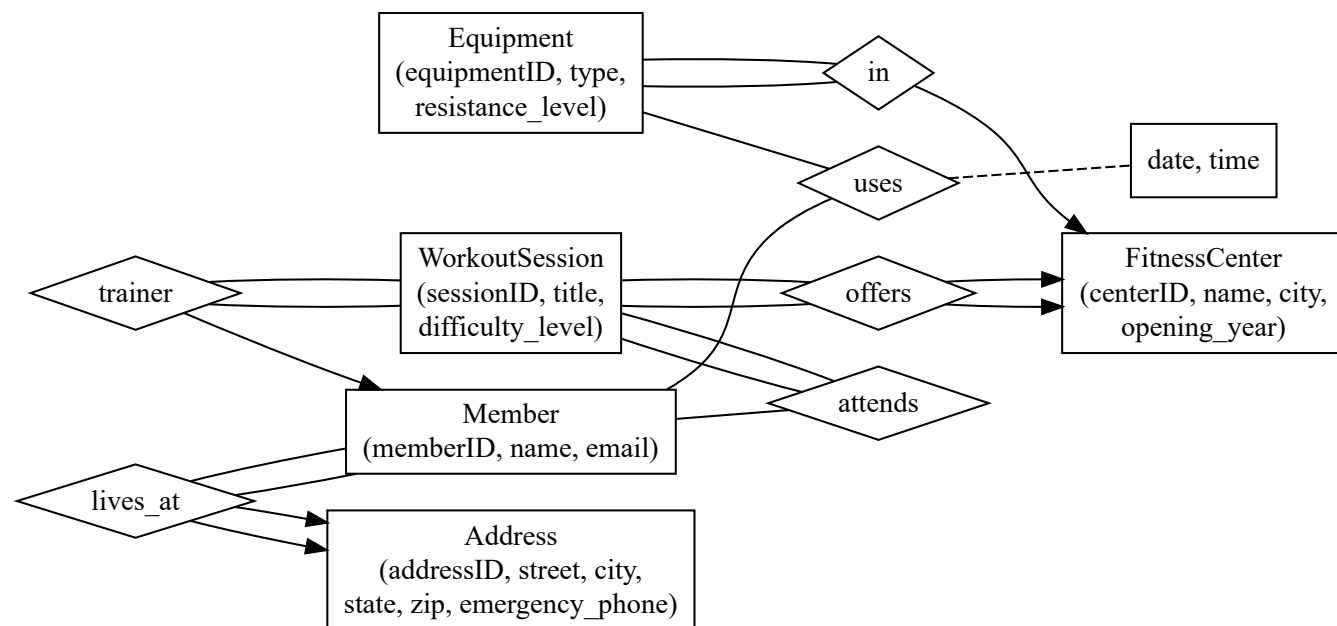
In Account and Loan, B_name should have a foreign key constraint and a not null constraint

In Depositor, Account_No and C_name should have foreign key constraints

In Borrower, Loan_No and C_name should have foreign key constraints

2

a



Assumptions

1. All Equipments must be in a Fitness Center
2. All Fitness Centers must offer at least one Workout Session
3. All Workout Sessions must have one trainer, at least one member attending, and be in one fitness center
4. All Members have one Address
5. All Addresses correspond to at least one Member
6. Member can use a piece of Equipment more than once so a date and time was created

b

Equipment(equipmentID, type, resistance_level, centerID)

FitnessCenter(centerID, name, city, opening_year, sessionIDs)

WorkoutSession(sessionID, title, difficulty_level, trainerID, attendee's IDs, centerID)

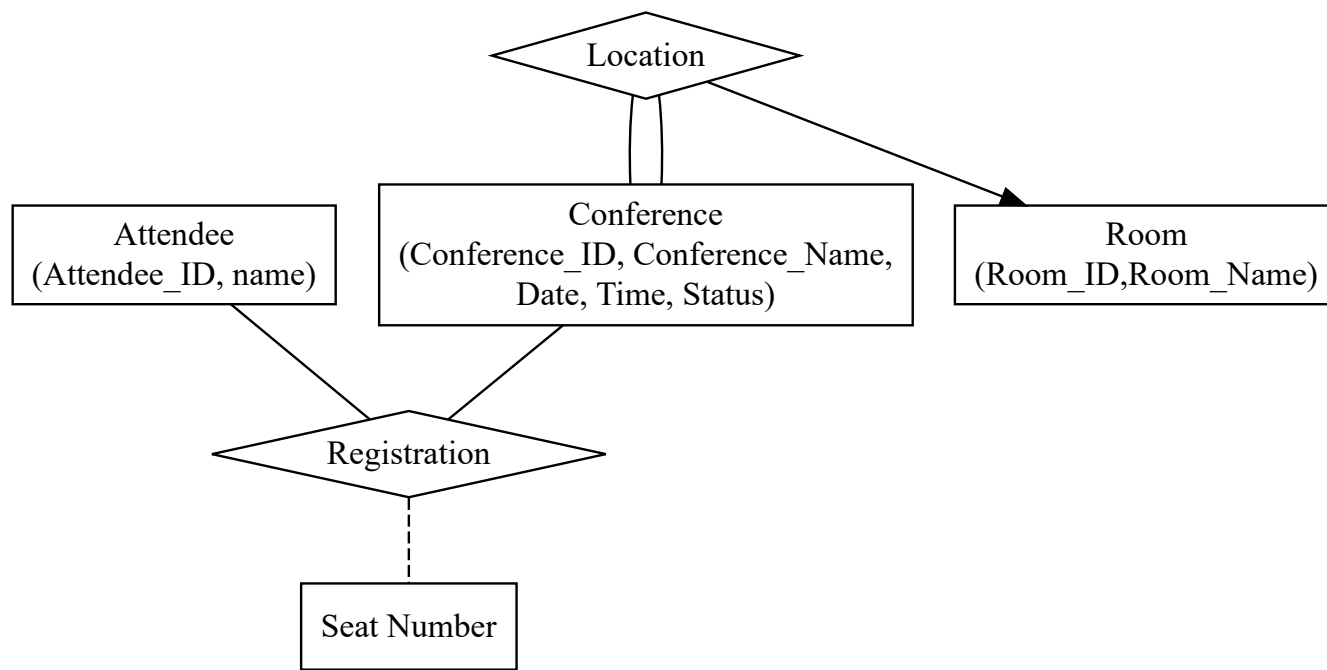
Member(memberID, name, email, addressID)

Address(addressID, street, city, state, zip, emergency_phone, memberIDs)

Uses(equipmentID, memberID, date,time)

3

a



Assumptions

1. Seats are numbered specifically for an event
2. Attendee only has one seat for an event
3. Conference only has one room
4. Conference_ID References a Unique instance of a conference (ie if a conference has multiple schedules they have different ids despite having the same name but different times)
5. Attendee doesn't need to be registired to a conference (could be in system from a previous year, maybe they didn't complete all registration steps etc)
6. Not all conferences need to have an attendee

b

Attendee(Attendee_ID, name)

Conference(Conference_ID, Conference_Name, Ddate, Time, Status, Room_ID, Room_Name)

Registration(Attendee_ID,Conference_ID, Seat Number)

Constraints

Registration should have foreign key constraints on Attendee_ID and Conference_ID and the two should have key constraints as well

Attendee and Conference should have key constraints on Attendee_ID and Conference_ID respectively